Code No: 113BP **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech II Year I Semester Examinations, September - 2021
DATA STRUCTURES
(Common to CSE, IT)

Time: 3 hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1. What is a Doubly-Linked List? What are the operations performed on it and write methods for each operation. [15]

2. Illustrate the stack operations in C using an array and explain the same for implementing recursive procedures. [15]

3. What is a Binary Tree? Explain Binary Tree Traversals. [15]

4. Write an algorithm for Quick Sort and explain with an illustrative example. [15]

5. What are Red-Black Tree? How to construct and insert elements in a Red-Black Tree? Show each step of the Red-Black Tree, built from a sequence of insertions corresponding to the following keys: 44,17,32,78,50,54,62,48,88. [15]

6. Write the algorithms for deletion of an element and insertion of elements in a Circular Linked List and explain the procedure of the same. [15]

7.a) What is a Double ended Queue? State various operations on Circular Queues.
b) Convert to postfix and evaluate the following expression using a stack: 2+3*4+6/2. [7+8]

8. What is a Graph? How is it represented? Explain different Graph Traversals. [15]

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Used Papers SEP 2021